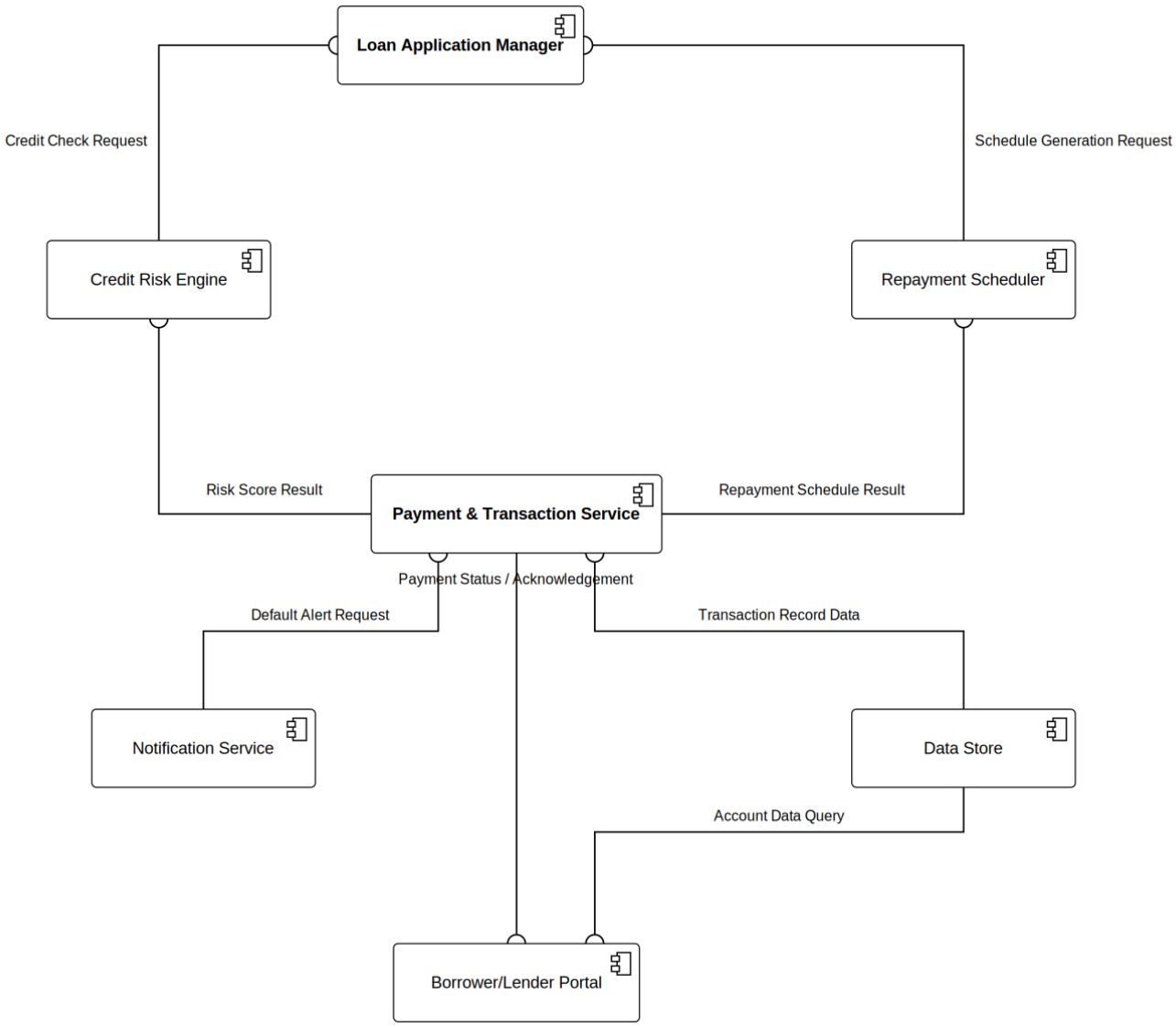
Lab 3: Software Architecture

Micro-Lending & Peer Credit Risk Assessor

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UML Component Diagram



Architecture Justification

Architectural Choice :

Selected Architecture: Layered Architecture

We selected Layered Architecture for the Micro-Lending & Peer Credit Risk Assessor system. The system is divided into separate components responsible for managing loan applications, assessing borrower credit risk, generating repayment schedules, processing payments, sending notifications, storing financial data, and displaying account information to borrowers and lenders.

Reason 1 — Separation of Concerns :

Layered Architecture separates different responsibilities of the system into independent components. The Loan Application Manager, Credit Risk Engine, and Repayment Scheduler handle loan evaluation and scheduling activities, while the Payment & Transaction Service handles EMI distribution and repayment processing. The Notification Service, Data Store, and Borrower/Lender Portal handle alerting, persistent storage, and reporting respectively. This separation makes the system easier to understand, maintain, test, and modify.

Reason 2 — Maintainability and Scalability :

The architecture allows individual components to be modified or extended without significantly affecting the rest of the system. For example, a new risk-scoring rule can be added to the Credit Risk Engine, or an additional repayment plan type can be introduced in the Repayment Scheduler, without redesigning the complete system. This makes the architecture suitable for future expansion of the lending platform.

Security Advantage :

Layered Architecture improves security by controlling how components communicate with each other. Borrowers and lenders interact with the Borrower/Lender Portal rather than directly accessing the Data Store or the Payment & Transaction Service. This separation helps protect sensitive financial information and limits direct access to components that execute monetary transactions.

Performance Benefit :

The architecture allows loan-processing operations to run independently from portal access. The Loan Application Manager can initiate credit risk assessment and schedule generation while the Borrower/Lender Portal retrieves already-stored account and repayment information from the Data Store. This reduces unnecessary repeated computation and allows users to view account information without waiting for a new risk assessment or scheduling operation.